INCH-POUND

MIL-G-82814(OS)
29 December 1989
SUPERSEDING
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15 July 1986

MILITARY SPECIFICATION

GRAPHITE REINFORCEMENT

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for one type of graphite reinforcement, referred to herein as graphite.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

STANDARDS

MILITARY

MIL-STD-129 Marking for Shipment and Storage

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, MD 20640-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 9330

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(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from: Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094).

2.1.2 **Other Government documents, drawings, and publications.** The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 100-199 Transportation

(Application for copies of CFRs should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402-0001.)

2.2 **Non-Government publications.** The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM C 604 True Specific Gravity of Solid Refractory Materials by Gas Comparison Pycnometer

ASTM D 1682 Breaking Load and Elongation of Textile Fabrics

ASTM D 1777 Measuring Thickness of Textile Materials

ASTM D 3176 Ultimate Analysis of Coal and Coke

ASTM D 3776 Weight (Mass) per Unit Area of Woven Fabric

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

AMERICAN TRUCKING ASSOCIATION, INC.

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, Inc., Attn: Traffic Dept., 2200 Mill Road, Alexandria, VA 22314-4677.)

NATIONAL RAILROAD FREIGHT COMMITTEE

Uniform Freight Classification (UFC) 6000

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(Application for copies should be addressed to the National Railroad Freight Committee, 222 South Riverside Plaza, Suite 1120, Chicago, IL 60606-5945.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 **Order of precedence.** In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 **Material.** The material shall be a square weave, rayon-base graphite fabric final processed at a temperature of 4200 to 5200°F with the properties defined in table I.

TABLE I. Properties.

Property	Limits	
	Minimum	Maximum
Carbon, weight percent	99.0	---
Ash content, weight percent	0	0.5
Moisture content, percent	0	0.5
Weight, ounce per square yard	6.5	8.5
Breaking strength, pound per inch of width		
Warp direction	10	---
Fill direction	10	---
Specific gravity at 25°C	1.38	1.50
Thickness, inch	0.017	0.027

3.2 **Workmanship.** The graphite shall be free from tear , holes, impurities, non-uniform width and thickness, and other defects which could render it unsuitable for its intended use.

4. QUALITY ASSURANCE PROVISIONS

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4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 **Responsibility for compliance.** All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 **Quality conformance inspection.** Quality conformance inspection shall consist of the tests specified in table II.

TABLE II. Quality conformance inspection.

Examinations and Tests	Requirement	Test Method
Visual examination	3.2 and Section 5	4.3.1
Carbon assay and moisture content	3.1	4.3.2
Ash content	3.1	4.3.3
Weight	3.1	4.3.4
Breaking strength	3.1	4.3.5
Specific gravity	3.1	4.3.6
Thicknes	3.1	4.3.7

4.2.1 **Lot.** A lot shall consist of all the graphite reinforcement rolls subjected to inspection at one time from one of the following:

a. A single batch (see 6.3.1)

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b. A continuous process (see 6.3.2)

4.2.2 **Sampling.** Except where 100 percent sampling is indicated, the number of rolls to be selected shall be in accordance with table III. Material shall be selected from the exposed end of each sampled roll in sufficient quantity to perform all the tests specified herein. Except as otherwise specified, three specimens shall be tested from each sample for each of the properties requiring test specimens. The average test results of three test specimens shall be reported in addition to the reporting of individual test specimen values.

TABLE III. **Sampling.**

Number of rolls in lot	Number of sample rolls selected for testing
1 to 3	All
4 to 40	3
41 to 65	4

4.3 **Inspection methods.** The following inspection methods shall be used. Unless otherwise specified in the applicable test method, all weights, volumes, temperatures, and times shall be measured to the nearest specified unit or decimal.

NOTE: Reagent grade chemicals shall be used for chemical reactions in the conduct of all tests defined herein. Solvents and indicators may be commercial nonreagent grade materials.

4.3.1 **Visual examination.** Visual examination shall be made to verify conformance to the requirements of 3.2 and section 5.

4.3.2 **Carbon assay and moisture content.** A carbon assay and moisture content determination shall be performed as follows:

a. Specimen shall be dried to constant weight at 225 ± 5°F before combustion. Percent carbon, to the nearest 0.1 percent, shall be based on the weight of the dried specimen.

b. Carbon assay shall be conducted in accordance with ASTM D 3176.

c. Determine moisture content as follows:

$$\text{Percent moisture} = \frac{W - W1}{W} \times 100$$

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Where: W = specimen weight prior to drying, g

W1 = specimen weight after drying, g

d. Record moisture content to the nearest 0.1 percent.

4.3.3 **Ash content.** The ash content shall be determined as follows:

a. Dry 3.0 ± 0.6 g of graphite fabric in a circulating oven at 225 ± 5°F for a minimum of 1 hour.

b. Remove specimen and place in a desiccator to cool.

c. Weigh 2.0 ± 0.4 g (W) of cooled specimen into a tared crucible (B) and place into a muffle furnace at 1560 ± 25°F for a minimum of 16 hours. Weights (W) and (B) shall be measured to the nearest 0.001 gram.

d. Allow an additional 30 minutes burning time or until constant weight is obtained if ashing is incomplete.

e. Remove specimen and crucible, place in a desiccator, and cool to ambient temperature.

f. Obtain post-heat weight of specimen and crucible (A) to the nearest 0.001 g and calculate percent ash as follows:

$$\text{Percent ash} = \frac{(A - B)}{W} \times 100$$

Where: A = weight of crucible and ash, g

B = weight of crucible, g

W = weight of specimen, g

g. Record ash content to the nearest 0.1 percent.

4.3.4 **Weight.** The weight of the material shall be determined in accordance with ASTM D 3776.

4.3.5 **Breaking strength.** The breaking strength test shall be conducted in accordance with ASTM D 1682.

4.3.6 **Specific gravity.** Specific gravity shall be determined in accordance with ASTM C 604.

4.3.7 **Thickness.** The thickness of the material shall be determined in accordance with ASTM D 1777 at 5 ± 4 pounds per square inch (psi).

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4.4 **Inspection of packaging.** The graphite containers and container markings shall be visually examined to verify compliance with Section 5.

5. PACKAGING

5.1 **Packaging.** Unless otherwise specified in the contract or order (see 6.2), packaging shall be level C as specified herein.

5.1.1 **Level C.** The graphite shall be packaged to afford adequate protection against loss, contamination, deterioration, and damage during shipment from the supply source to the first receiving activity. Containers in the same shipment shall be of the same size. The packaging shall conform to UFC 6000, National Motor Freight Classification, 49 CFR 171-178, or to other carrier rules and regulations as applicable to the mode of transportation.

5.2 **Marking.**

5.2.1 **Standard marking.** In addition to any special marking required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129.

5.2.2 **Special marking.** In addition to the marking requirements of 5.2.1, each container marking shall include the following:

a. Title, number, and date of this specification.

b. Manufacturer's name and address.

c. Net weight of contents and number of rolls.

d. Container identification number.

e. Purchase document number.

f. Contractor's name, lot number, and material name on inside of spool.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 **Intended use.** The graphite is intended for use in graphite cloth phenolic for reinforced plastic nozzle components.

6.2 **Acquisition requirements.** Acquisition documents must specify the following:

a. Title, number and date of this specification.

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b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.1.2).

c. Special marking, if other than as specified in 5.2.2

6.3 **Definitions.**

6.3.1 **Batch.** A batch is that amount of graphite made as one unit in the manufacturing process.

6.3.2 **Continuous process.** A continuous process is an uninterrupted process of manufacture from which, in one unbroken flow, the graphite is accumulated for a lot.

6.3.3 **Roll.** A single roll of continuous or joined cloth delivered to the impregnator.

6.3.4 **Sample.** A roll of cloth selected in accordance with the requirements of this specification for performance of inspections.

6.3.5 **Specimen.** A portion of material taken from a sample roll for performance of inspections.

6.4 **Subject term (key word) listing.**

Rocket Motor, Dual Thrust, Mk 104
Standard Missile

Preparing Activity:
NAVY-OS
(Project 9330-N023)

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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-R-82814	2. DOCUMENT DATE (YYMMDD) 29 December 1989

3. DOCUMENT TITLE
REINFORCEMENT, GRAPHITE

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)
	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY NAVAL ORDNANCE STATION (CODE 3730) INDIAN HEAD, MD 20640-5000

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
c. ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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